NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES

CS-4063 — Natural Language Processing

Assignment-02

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Overview

This assignment is about building a NLP pipeline for BBC Urdu text data using PyTorch from scratch. This is the continuation of the previous assignment.

Part 1 focuses on word representation methods including TF-IDF, PPMI and Skip-Gram Word2Vec and then evaluates them with nearest neighbour behaviour, analogy style probing, and mean reciprocal rank.

Part 2 builds BiLSTM models from scratch for POS and NER under frozen and fine tuned embedding settings, then evaluates both token level and entity level behavior.

Part 3 implements a transformer encoder from scratch, including custom scaled dot product attention, multi head self-attention, positional encoding, pre layer normalization blocks, and a CLS based classifier for five-way topic prediction.

Part 1

Part 1 compares four embedding conditions and uses a consistent evaluation recipe to make the comparison fair. The MRR summary indicates the following ranking: C1 PPMI baseline achieved 0.1478 and was the best condition, C2 skip gram on raw text achieved 0.0995, C3 skip gram on cleaned text also achieved 0.0995, and C4 skip gram on cleaned text with d equals 200 achieved 0.0310. This ranking suggests that for this corpus and evaluation pair set, explicit co-occurrence weighting with PPMI provided stronger retrieval behaviour than the learned skip gram settings. The nearest neighbour and analogy outputs were only partially coherent, which is expected for a relatively small and noisy corpus with many domain specific or low frequency tokens. Another important observation is that increasing embedding dimension alone did not improve ranking quality here, the d=200 condition performed worst by MRR, likely due to data sparsity and over parameterization relative to corpus size.

Overall, Part 1 establishes a useful practical baseline and motivates why later model components should be evaluated with task specific metrics rather than relying only on intrinsic embedding checks.

Part 2

Part 2 has two distinct outcomes. For POS tagging, the BiLSTM setup performed well and improved notably when embeddings were fine tuned. Frozen embeddings produced test accuracy of 0.8527 and macro F1 of 0.7301, while fine tuned embeddings produced test accuracy of 0.9556 and macro F1 of 0.8103. This gain confirms that adapting embeddings to the supervised sequence objective helps when token level labels are sufficiently frequent and consistent. The confusion patterns also indicate that most frequent tags are learned robustly, while minority or lexically ambiguous tags remain harder. For NER, however, both CRF and

non CRF variants reported zero precision, recall, and F1 at entity level in test evaluation, in both frozen and fine tuned modes. The validation entity F1 stayed at 0.0000 across epochs for the non CRF runs, and CRF test reports also showed all zeros per type and overall. This is not a scoring bug by itself, it indicates prediction collapse toward O labels or boundary mismatches under strict conlleval style matching. The split statistics show that O dominates while some entity types have very small support, making optimization unstable and strongly biased toward the majority class.

In short, Part 2 demonstrates that the same BiLSTM backbone can succeed on dense token supervision (POS) yet fail under sparse span supervision (NER) without stronger balancing, augmentation, or label quality controls.

Part 3

Part 3 implements a custom transformer encoder stack without using built in transformer classes and evaluates it on five class topic classification. The model was trained with AdamW, cosine schedule, and warmup, then evaluated on held out test data. The final test accuracy was 0.9901 and test macro F1 was 0.9501, with best validation accuracy reaching 0.9959. The 5 by 5 confusion matrix shows very strong diagonal concentration and limited confusion across classes. Representative counts include Politics correctly predicted 82 with a few errors toward Health and Society, Sports correctly predicted 197 with very few off diagonal errors, Economy correctly predicted 18 with one error toward International, International correctly predicted 26 with a few errors toward Health and Society, and Health and Society correctly predicted 1872 with one error toward International. Attention heatmaps from two heads in the final encoder layer for correctly classified articles reveal non uniform focus: some heads place high weight on CLS to aggregate global evidence, while others emphasize a small set of content words likely carrying topical cues.

The head diversity supports the interpretation that different heads capture complementary token relations. Compared with BiLSTM topic style baselines discussed in the notebook, the transformer gave higher classification performance in this setup, though with heavier per epoch compute due to pairwise attention operations.

Conclusion

The assignment demonstrates a complete end to end NLP workflow, from representation learning to structured prediction and transformer classification, with reproducible artifacts and clear empirical trends. First, intrinsic embedding quality was moderate and condition dependent, with PPMI outperforming skip gram variants on the selected MRR probe. Second, supervised sequence learning with BiLSTM was effective for POS but not for NER under current label density and evaluation strictness, highlighting the mismatch between decreasing

training loss and usable entity extraction quality. Third, the custom transformer encoder delivered excellent topic classification results, both in overall accuracy and macro F1, and produced interpretable attention patterns that align with keyword driven topic decisions. A practical takeaway is that architecture choice must be tied to supervision regime and dataset scale: recurrent models can be robust and efficient for dense token labels, while attention based models can excel on document level categorization when enough discriminative lexical signals are present. For future improvements, NER should be prioritized with stronger annotation quality, class rebalancing, and objective design changes such as weighted losses or focal variants, while maintaining strict held out evaluation to verify genuine entity span recovery.